

Differential Geometry and Lie Groups

Semestral Examination

November 17, 2025

Instructions: All questions carry ten marks. Manifolds are assumed to be smooth.

1. Let M be a manifold and X be a smooth vector field on M . Define integral curve for X . Prove that the domain of the integral curve of a left-invariant vector field X on a Lie Group G is the whole of the real line (i.e. any such X is a *complete* vector field).
2. Define *involutive distribution* on a manifold M . Give an example of a distribution that is not involutive.
3. Define Riemannian metric on a manifold M . Prove that Riemannian metric exists on any manifold M .
4. Define a C^∞ -connection ∇ on M . If X, Y are smooth vector fields, then for any $p \in M$, prove that $\nabla_X Y|_p$ depends only on values of X and Y in an open neighbourhood of p .
5. Define the curvature tensor on M . If S is a surface in $\mathbb{R}^3$, then express the Gaussian curvature of S at a point p in terms of the curvature tensor of the Riemannian connection associated with its First Fundamental form.
6. Define sectional curvature of a manifold M at a point p . Prove that the Riemannian curvature at a point is uniquely determined if we know all sectional curvatures at that point.